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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

MAKE UP EXAMINATIONS – OCTOBER 2023

Program	:	**B.E. - Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Database Management Systems**	**Max. Marks**	:	**100**
Course Code	:	**IS45**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Discuss the characteristics of a database approach..	CO1	(08)
	b)	When would it make sense not to use a database system? Explain with one example.	CO1	(04)
	c)	A Company database needs to store information about employees who have unique SSN, salary and phone number. It also stores information about department having details like dno, dname and budget, and also children with details name and age. In this database every employee should work in any one department, each department is managed by an employee; a child must be identified uniquely by the name. Draw an ER diagram corresponding to the above scenario.	CO1	(08)
2.	a)	Explain the different phases of Database design with a neat diagram.	CO1	(08)
	b)	Illustrate Database Schema and Instance with an example for each.	CO1	(04)
	c)	A university database contains information about Professors, who are identified by social security number or SSN and Courses has course id. Professors teach courses; each of the following situations concerns the Teaches relationship set. For each situation, draw an ER diagram that describes it. i. Professors can teach the same course in several semesters, and each offering must be recorded. ii. Professors can teach the same course in several semesters, and only the most recent such offering needs to be recorded. (Assume this condition applies in all subsequent questions.) iii. Every professor must teach some course. iv. Every professor teaches exactly one course. v. Every professor teaches exactly one course, and every course must be taught by some professor.	CO1	(08)

UNIT - II

3.	a)	List the characteristics of relations. Explain any two.	CO2	(06)
	b)	Discuss the various reasons that lead to the occurrence of NULL values in relations.	CO2	(06)
	c)	Consider the following COMPANY database EMP(Name,<u>SSN</u>,Salary,SuperSSN,Gender,Dno) DEPT(<u>DNum</u>,Dname,MgrSSN,Dno) DEPT_LOC(<u>Dnum,Dlocation</u>) DEPENDENT(<u>ESSN,Dep_name</u>,Sex) WORKS_ON(<u>ESSN,Pno</u>,Hours) PROJECT(Pname,<u>Pnumber</u>,Plocation,Dnum)	CO2	(08)

Write the relational algebra queries for the following:
(i) Retrieve the name, address, salary of employees who work for the Research department.
(ii) Find the names of employees who work on all projects controlled by department number4.
(iii) Retrieve the names of employees who have no dependents
(iv) Retrieve each department number, the number of employees in the department and their average salary.

4.	a)	Discuss the entity integrity and referential integrity constraints. Why is each considered important?	CO2	(05)
	b)	Consider the following GRADEBOOK relational schema describing the data for a grade book of a particular instructor. (Note: The attributes A, B, C, and D of COURSES store grade cutoffs.) CATALOG(<u>Cno</u>, Ctitle) STUDENTS(<u>Sid</u>, Fname, Lname, Minit) COURSES(<u>Term, Sec_no</u>, Cno, A, B, C, D) ENROLLS(<u>Sid, Term, Sec_no</u>) Specify and execute the following queries using the Relational Algebra on the GRADEBOOK database schema. i. Retrieve the names of students enrolled in the Automata class during the fall 2009 term. ii. Retrieve the Sid values of students who have enrolled in CSc226 and CSc227. iii. Retrieve the Sid values of students who have enrolled in CSc226 or CSc227. iv. Retrieve the names of students who have not enrolled in any class. v. Retrieve the names of students who have enrolled in all courses in the CATALOG table.	CO2	(10)
	c)	Draw Relational schema for the above Question 4.b) with Referential integrity constraints and specify the foreign keys.	CO2	(05)

UNIT - III

5.	a)	Write SQL syntax for the following with example: (i) SELECT (ii) ALTER (iii) UPDATE.	CO4	(06)
	b)	List the data types that are allowed for SQL attributes.	CO4	(06)
	c)	Consider the following relation schema Works(Pname,Cname,salary) Lives(Pname,Street,City) located_in (Cname, city) Manager(Pname,Mgrname) Write the SQL queries for the following i) Find the names of all persons who live in the city Bangalore. ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000 iii) List the names of the people who work for “Tech M” along with the cities they live in. iv) Find the average salary of “Infosys” persons.	CO4	(08)
6.	a)	Describe the six clauses in the syntax of an SQL retrieval query. Show what type of constructs can be specified in each of the six clauses. Which of the six clauses are required and which are optional?	CO4	(10)

b) Consider the following schema of Company database. Write the SQL queries for the following. CO4 (10)

EMP(Name,SSN,Salary,SuperSSN,Gender,Dno)
DEPT(DNum,Dname,MgrSSN,Dno)
DEPT_LOC(Dnum,Dlocation)
DEPENDENT(ESSN,Dep_name,Sex)
WORKS_ON(ESSN,Pno,Hours)
PROJECT(Pname,Pnumber,Plocation,Dnum)

i) Retrieve the name of each employee who works on all the projects controlled by department number 5 (use non-Exists)
ii) Retrieve the Social Security numbers of all employees who work on project numbers 1, 2, or 3
iii) Retrieves the name and address of every employee who works for the 'Research' department
iv) Find the sum of the salaries of all employees of the 'Research' department, as well as the maximum salary, the minimum salary, and the average salary in this department.
v) For each department, retrieve the department number, the number of employees in the department, and their average salary.

UNIT- IV

7. a) List all the Informal design guidelines for relational database design. CO4 (06)
b) Define Functional dependency. Discuss the inference rules for Functional dependency. CO4 (08)
c) Define BCNF. How does it differ from 3NF? Why is it considered a stronger form of 3NF? CO4 (06)

8. a) Why Normalization is important? When you say that the relation schema satisfies First Normal form. Give an example. CO4 (08)
b) Given a relation R (A, B, C, D) and Functional Dependency set FD = {AB → CD, B→ C}, determine whether the given R is in 2NF? If not convert it into 2 NF. CO4 (08)
c) Discuss the problem of spurious tuples and how we may prevent it. CO4 (04)

UNIT - V

9. a) What do you mean by State Transition diagram? When is a transaction said to be failed? Explain with a diagram. CO5 (06)
b) What is ARIES? Discuss its three phases. On what principles is it based on? CO5 (08)
c) Discuss the various data types that the MongoDB supports. Show with an appropriate example for the use of those data types. CO5 (06)

10. a) Discuss shadow paging. How it is different from log based recovery techniques? Mention some of it disadvantages. CO5 (07)
b) What is the system log used for and list the typical kinds of records in a system log? What are the transaction commit points? CO5 (07)
c) What are the characteristics of a NoSQL database? Discuss any two types of NoSql Databases. CO5 (06)

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